Erik Jin
Student ID: 21431993

Problem 1:

$T = 5$

Face Value = 1200
$C = 5\%$
$m = 2$ (payable semi-annually)
$y = 4\%$ annual effective rate

a) The current price of the bond is the present value of future cash flow. We use the yield as discount rate.

$$P_0 = \sum_{i=1}^{N} \frac{\text{Face value} \cdot C/m}{(1+y)^{i/m}} + \frac{\text{Face Value}}{(1+y)^T} = 1256.07$$

where $N = T \cdot m$ is the number of coupon payments

b) The formula for the modified duration and modified convexity

$$D_{mod}(i) = -\frac{P'(i)}{P(i)} = \frac{D_{mac}}{1+i} \quad \text{where } D_{mac} = \frac{\sum_k t_k C_k (1+i)^{-t_k}}{\sum_k C_k (1+i)^{-t_k}}$$

Where t_k and C_k denotes the time and payment amount at time k

$$D_{mod}(y) = 4.33$$

$$C_{mod} = \frac{P''(i)}{P(i)} = \frac{\sum_k t_k (t_k+1)(1+i)^{-(t_k+2)}}{\sum_k C_k (1+i)^{-t_k}}$$

$$C_{mod}(y) = 24.13$$

c) $\varepsilon = -0.1\%$

The percentage change in bond price using second order approximation

$$\frac{P(i_0+\varepsilon) - P(i_0)}{P(i_0)} = -\varepsilon D_{mod} + \frac{\varepsilon^2}{2} C_{mod} = \{i_0 = y, \varepsilon = -0.1\%, D_{mod} = 4.33, C_{mod} = 24.1$$

$$= 0.43\%$$

Problem 2

3-year coupon bond

$F = 1000$

$C = 4\%$

$m = 2$

$T = 3$

a) The yield of the bond can be viewed as the discount rate that makes the Net Present Value of buying the bond equal to 0

$$0 = -P_0 + \sum_{k=1}^{N=T\cdot m} \frac{F\cdot C/m}{(1+y)^{k/m}} + \frac{F}{(1+y)^T}$$

We want to determine y such that the equation above holds. This can be transformed {with $u = \frac{1}{(1+y)^{1/2}}$} as a degree 6 polynomial in u.

Instead of trying to solve this analytically we can implement root solving algorithms such as brentq from Scipy.optimize.

The initial price is calculated using the spot rates as discount rates. This can be motivated as the cash flows can be replicated with zero coupon bonds where the yield make up the term structure that describe the spot rates.

$$P_0 = \sum_{i=0}^{N} \frac{F\cdot C/m}{(1+S_0(i/m))^{i/m}} + \frac{F}{(1+S_0(T))^T}$$

$P_0 = 1026{,}98$

Using the root solving algorithm we get

$y_{eff} = 3{,}0754\%$ This is quoted as annual effective yield

We can covert it into an annual nominal yield (YTM)

with $y_{nom} = \left((1+y_{eff})^{1/m} - 1\right)\cdot m = 3{,}0522\%$

b) Here we are asked to calculate the first order approximation of the percentage change of bond price when the annual nominal yield is increased by 0,2%

This can be done with

$$\frac{P(i_0+\varepsilon)-P(i_0)}{P(i_0)} = -\varepsilon \frac{D_{mac}}{1+i_0}$$

The Macaulay duration $\frac{\sum_k t_k C_k (1+i)^{-t_k}}{\sum_k C_k (1+i)^{-t_k}}$

is the same if you use annual effective or annual nominal! However the discount factor uses effective rate

When we convert from annual nominal to effective we use $j = \frac{i^{(m)}}{m}$ where $i^{(m)}$ is the annual nominal rate and j is the effective rate.

Using $i^{(m)} = y_{nom}$ from a) and $\varepsilon = 0,2\%$ we get

$$\frac{P(i_j^{(m)}+\varepsilon)-P(i^{(m)})}{P(i^{(m)})} = -0,5631\%$$

Problem 3

3-year floating rate note with

$F=100$

$m=2$

$C=\{6\text{-month spot rate}\}+0,25\%$

We are given spot rate from $t=6$ months to $t=3$ years with 6 month intervals.

a) We can calculate the price of the bond using the general principle that two portfolios with identical cash flows must have the same price, or setup cost, if the market is free of arbitrage.

We can create a replicating portfolio and consider the setup cost.

The coupons can be decomposed into a pure floating rate bond and an annuity that pays 0,25% Face Value every 6 months.

The pure floating rate note has a cost that is equal to the face value after a coupon has been paid or right at inception.

The annuities can be priced by replicating the payments with zero coupon bonds with Face value $=0,25\%\,F$ and maturity iT_c for $i=1,\dots,6$ where $T_c=6$ months

The yield of these are given by the spot rates.

Thus, the price of the 3-year coupon bond is

$$P_0=F+\sum_{i=1}^{N} 0,25\%\,F\,B(0,iT_c)=101,44$$

Problem 3

b) After 9 months the spot rates have changed

We are given that the spot rate at the 1st coupon payment date ($t=6$ months, 3 months ago) is 2% (annual effective)

To calculate the price of the bond at $t=9$ months we need to calculate the size of the coupon payable at $t=12$m as well as the price of the floating rate note at $t=12$ months. and the discounted annuities.

Then we can discount it to $t=9$ months using the 3-month zero coupon bond.

The coupon is based on the 6-month spot rate from the last coupon payment period. The coupon is calculated using the effective rate. The conversion from annual effective to effective is calculated with

$$j = (1+i)^{1/m} - 1$$

Thus the first coupon after 9 months is

$$CF = \left((1+2\%)^{1/2} - 1\right) \cdot 100$$

The annuities can be replicated with

$\sum_{i=1}^{N-1} 0{,}25\% \, F \, B(0, i T_c - \frac{3}{12})$, where we shift with 3 months as that is the amount of time that has passed since the last annuity

At month 12 after the coupon has been paid the floating rate note (pure part) is worth the face value

Putting the three components together we get

$$P_{9m} = \underbrace{\sum_{i=1}^{N-1} 0{,}25\% \, F \, B(0, i T_c - \frac{3}{12})}_{\text{PV annuity}} + \underbrace{\left[F + \left((1+i_6)^{1/2} - 1\right) F\right]}_{\text{value of FRN just before coupon payment}} \underbrace{B\left(0, \frac{3}{12}\right)}_{PV} = 101{,}76$$

Problem 4

An investor has a short position of a 2-year contract on a 5-year coupon bond with

$F = 1000$

$C = 5\%$

$m = 2$

$y_0 = 5\%$ the current annual effective yield of the bond

We are also given spot rates of maturities

6, 12, 18 and 24 months.

After 1 year the spot rates decrease uniformly by 0,5% and the yield of the bond drops to $4,5\% = y_1$

The investor can "cancel" the short position with a trading strategy. The investor needs to construct a portfolio with the opposite payoff relative to the contract.

As the investor has a short position in the contract, the payoff at maturity ($t=2$) is

$X - S_2$ where X is the delivery price agreed at $t=0$ and S_2 is the value of the bond at $t=2$.

This payoff can be cancelled if the investor can create a portfolio with payoff $S_2 - X$ at $t=2$

At $t=1$ this can be achieved by entering in a long position in the bond, and short positions in three zero coupon bonds; one with face value and maturity $\left(\frac{CF}{m}, 6 \text{ months}\right)$, one with $\left(\frac{CF}{m}, 12 \text{ months}\right)$ and one with $(X, 12 \text{ months})$

The first two cancel out the coupons from the bond and the last one replicate $-X$ at $t=2$

Thus, the value of this portfolio is

$$S_1 - C\frac{F}{m}\left(B(0,1/2) + B(0,1)\right) - XB(0,1)$$

To calculate the cost of the strategy

We first need to calculate the initial cost of the bond

We use the present value of the cash flows

$$P_0 = \sum_{i=1}^{N} \frac{CF/m}{(1+y_0)^{i/m}} + \frac{F}{(1+y_0)^{T_B}} \quad \text{where } T_B = 5 \text{ years}$$

$P_0 = 1002,67$

The formula for a forward contract on a coupon paying bond is

$$f_0 + XB(0,T) = S_0 = \sum_{i=1}^{k} \frac{CF}{m} B(0,iT_c)$$

where $k = [T]$ $k = mT$: number of coupon payments before maturity of the contract T.

At inception the delivery price is set such that the value of the long position in the contract $f_0 = 0$

$$\Rightarrow X = \frac{1}{B(0,T)} \left(S_0 - \sum_{i=1}^{k} \frac{CF}{m} B(0,iT_c) \right) = 1003.39$$

We use this to calculate value of the short position of the contract $-f_1$ at time $t=1$ when conditions have changed

$$f_1 = S_1 - \sum_{i=1}^{k-2} \frac{CF}{m} B(0,i/m) - XB(0,1)$$

where $S_1 = \sum_{i=1}^{N-2} \frac{CF/m}{(1+y_1)^{i/m}} + \frac{F}{(1+y_1)^{T_B-1}} = 1019,93$

$f_1 = 7,5239$

$\Rightarrow$ The value of the short position is $-7,5239$

Thus the cost to cancel out the short position is $7,5239$

You can arrive at the same cost by replicating the payoff of the long position of the contract at $t=1$ (or enter a long position of an identical contract)

Bonus Problem

a) Current spot rates are

$\frac{1}{2}$m = 3,4%
1y = 3,5%
2y = 3,8%
all annual effective

The 18 month forward rate starting in 6 months is $f_{[0,5;2]} = 3,6\%$ (annual effective)

The implied 18 month forward rate can be calculated with

$$(1+R_{0,5})^{0,5}(1+f^{implied}_{[0,5;2]})^{(2-0,5)} = (1+R_2)^2$$

$$\Rightarrow f^{implied}_{[0,5;2]} = \left(\frac{(1+R_2)^2}{(1+R_{0,5})^{0,5}}\right)^{1/1,5} - 1 = 3,9337\%$$

The actual forward rate is lower than the implied $\Rightarrow$ there is an arbitrage opportunity

We want enter a forward contract on the rate between $t=0,5$ and $t_2=2$.

We long a 2-year zero coupon bond with face value N

We want to create a position where we pay N at t_2

This can be achived by ~~investing~~ in a shorting a 0,5-year zero coupon bond and refinancing it with a 1,5-year ~~zero coupon bond~~ loan. As we have the forward contract we pay at the rate $f_{[0,5;2]}$ instead of $R_{[0,5;2]}$. If we have to pay N at $t=2$ then we want to short a 0,5 year zero coupon bond with face value $\frac{N}{(1+f_{[0,5;2]})^{1,5}}$. The value at $t=0$ is

$\frac{N}{(1+f_{[0,5;2]})^{1,5}(1+R_{0,5})^{0,5}}$. At $t=0$ we get a risk free profit of $N\left(\frac{1}{(1+f_{[0,5;2]})^{1,5}(1+R_{0,5})^{0,5}} - \frac{1}{(1+R_2)^2}\right)$

$= 0,004488\,N$

b) We have an on going coupon bond with

$F = 1000$
$c = 5\%$
$m = 1$
$t = 9{,}5$ years
$S_0 = 1025$

The next coupon will be paid in 0,5 years

The 2-year forward price is currently 1000 =: X

We are given the spot rates with maturity 6, 12, 18 and 24 months

There is an arbitrage opportunity if the market implied forward price ≠ current forward price.

We calculate the implied forward price by replicating the payoffs when having a long position in the contract

$$f_0 + X_{imp} B(0,2) = S_0 - cF\left(B(0,0{,}5) + B(0,1{,}5)\right)$$

$f_0 = 0$ at inception $\Rightarrow$ $X_{imp} = \frac{1}{B(0,2)}\left(S_0 - cF\left(B(0,0{,}5) + B(0,1{,}5)\right)\right)$

Using the spot rates given

$X_{imp} = 1024{,}5569 > X$

The market forward price is undervalued $\Rightarrow$ We want to buy (long) it and short the portfolio that generates opposite payoff.

If we enter a long position in a forward contract on the bond the payoff at $t=2$ is $S_2 - X$

To create the opposite payoff we can buy a 2-year zero coupon bond with face value X and short the coupon bond that will give the payoff $-CF$ at $t=0{,}5$, $1{,}5$ and $-S_2$ at $t=2$

To cancel out the coupons we can long zero coupon bonds with face value CF at maturity 0,5 and 1,5. Thus we can construct a portfolio with

Long position: contract, $XB(0,2)$, $cFB(0,0{,}5)$ $cFB(0,1{,}5)$

Short position: S_0

The future cash flows are net 0 and setup cost is $f_0 + XB(0,2) + cF\left(B(0,0{,}5) + B(0,1{,}5)\right) - S_0$

The setup cost is $-22{,}2738 \Rightarrow$ Profit of 22,2738 at $t=0$ and no future cash flows